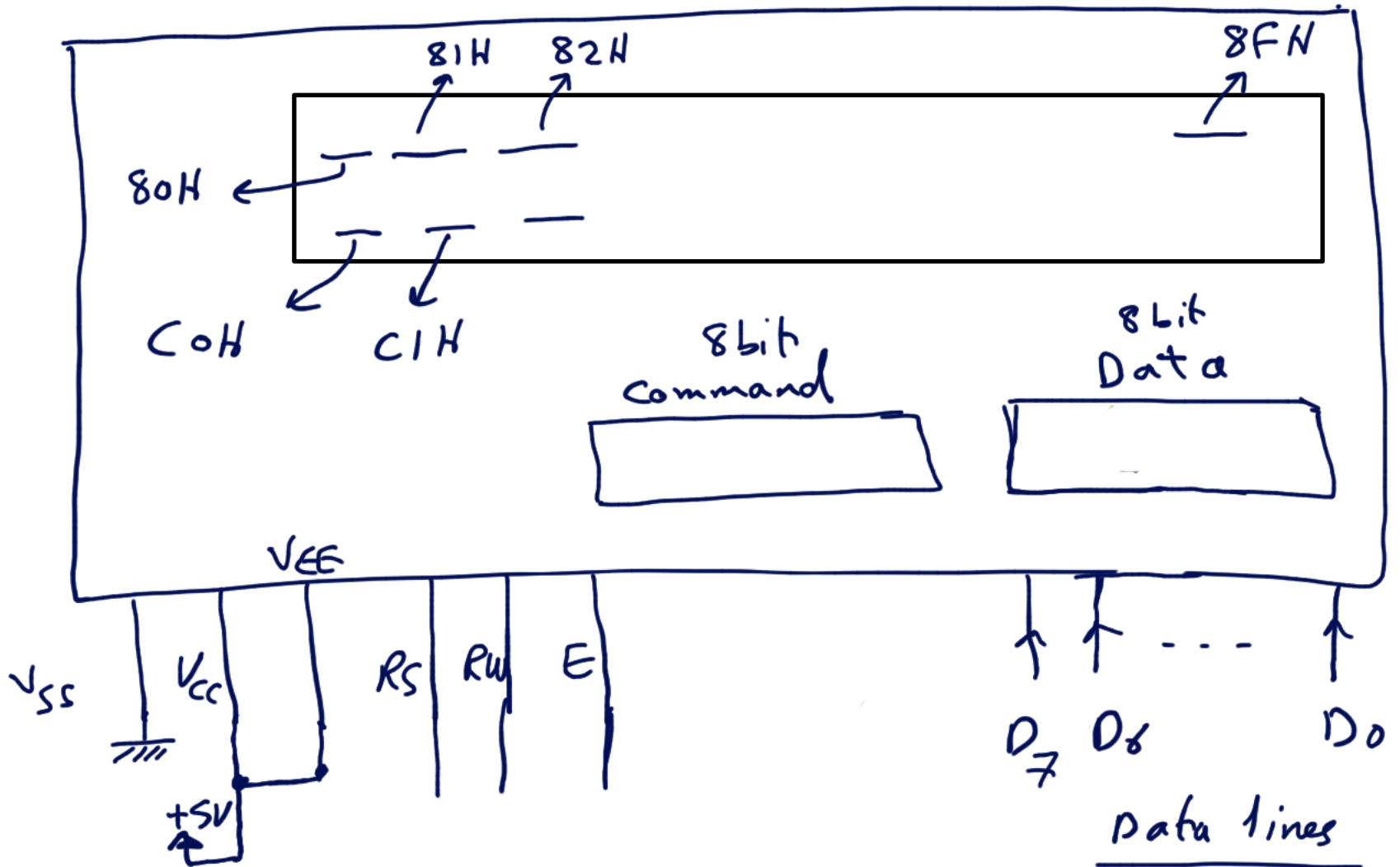


L#3 LCD & keyboard

LCD



V_{CC}: power supply

V_{EE}: contrast control

- E
- ① RS = 1 (data)
 - ② RW = 0 write to LCD
 - ③ E = H-to-L

RS: register select

②

RS = 0 select command register

RS = 1 select Data register

RW: (R/W) : Read / write

RW = 0 write data/command to LCD
(send)

RW = 1 read from LCD
(receive) $\overline{RD}/\overline{WL}$
(Ready)

E: enable



if RW = 0 $\rightarrow$ write $\rightarrow$ E = high-to-low pulse
(H-to-L)

if RW = 1 $\rightarrow$ read $\rightarrow$ E = low-to-high pulse
(L-to-H)

D₀ $\rightarrow$ D₇: Data lines are used to send/receive data/command from LCD.

Commands :

(3)

01H: Clear LCD

06H: shift cursor to the right.

08H: Display off, cursor off → Lab

0EH: Display on, cursor on

0FH: " on, " blinking (flash)

80H: force cursor to the beginning of 1st line.

C0H: " " " " " 2nd ".

38H: Define LCD, it has 2 lines, 5x7 matrix and 8 data lines.

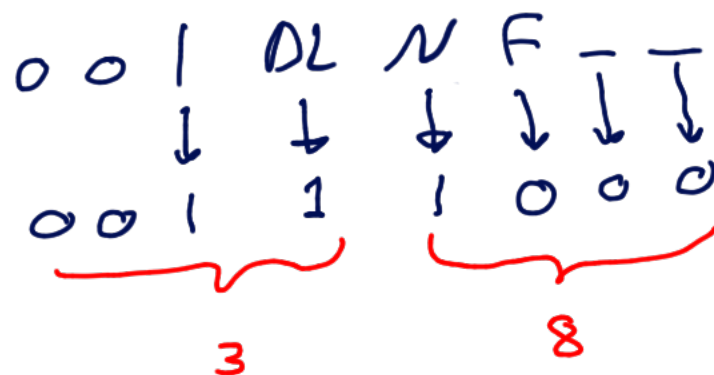


Function set:

F: Font

F=0 → 5x7 dots (matrix)

F=1 → 5x10 " "



No. # of lines

$N=0 \rightarrow 1 \text{ line}$

$N=1 \rightarrow 2 \text{ lines}$

DL : Data lines

$DL=0 \rightarrow 4 \text{ pins data lines}$

$DL=1 \rightarrow 8 \text{ pins " "}$

What is the command to configure LCD of 1 line
, 5x10 dots and 4 data lines

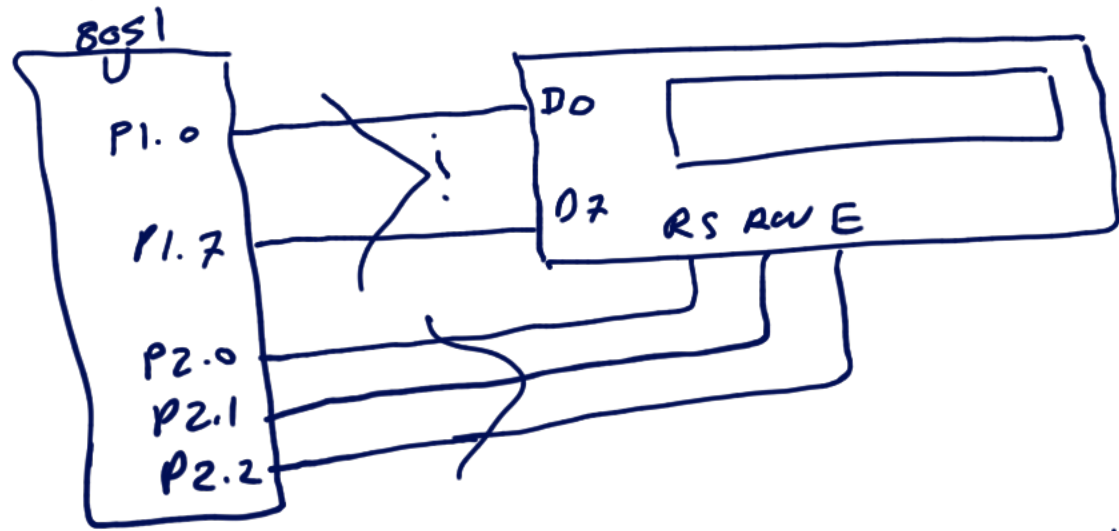
00 1 DL N F --

00 1 0 0 1 0 0 = 24H

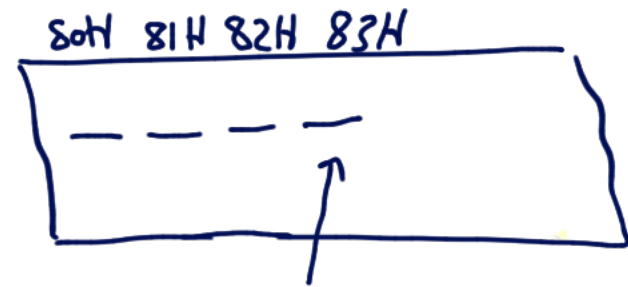
24H

Given the following interface

(3)



write a program to display your name on the
third position of first line of LCD. Then display
your ID on the second line of LCD.



6

Dline EQU P1
RS BIT P2.0
RW BIT P2.1
E BIT P2.2

ORG 0

initialize
LCD

MOV A, #38H

ACALL CMD

ACALL LDELAY

MOV A, #0EH

ACALL CMD

ACALL LDELAY

MOV A, #01H

ACALL CMD

ACALL LDELAY

} clear LCD

MOV A, #06

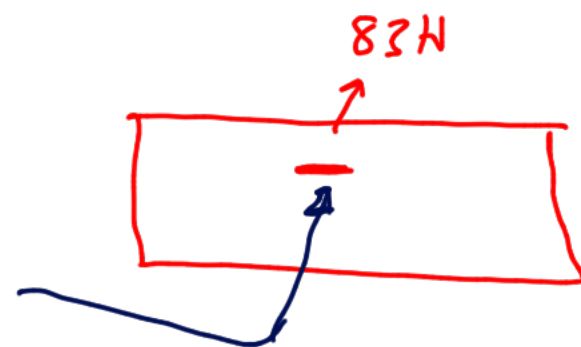
ACALL CMD

ACALL LDELAY

MOV A, #83H

ACALL CMD

ACALL LDELAY



7

MOV OPTR, #myname

ACALL USTR

MOV A, #0C0H

ACALL CMD

ACALL LDELAY

MOV OPTR, #myid

ACALL USTR

SJMP \$

Fadi

CMD: MOV DLine, A

CLR RS ; Command

CLR RW ; send (write)

SETB E

ACALL DELAY2 ; 

CLR E

RET

write
command
to LCD

WCHR: MOV DLine, A

SETB RS ; data

CLR RW

SETB E

ACALL DELAY2

CLR E

RET

write
Data
to LCD

WSTR: CLR A

MOVC A, @A+DPTR

CJNE A, #0, NEXT1

SJMP stop

NEXT1: ACALL WCHR

ACALL LDELAY

INC DPTR

SJMP WSTR

stop: RET

LDELAY: —
—
—
—

RET

DELAY2: —
—
—
RET

Write string
on LCD

myname: DB "Gadi", 0

myid: DB "ID: ...", 0

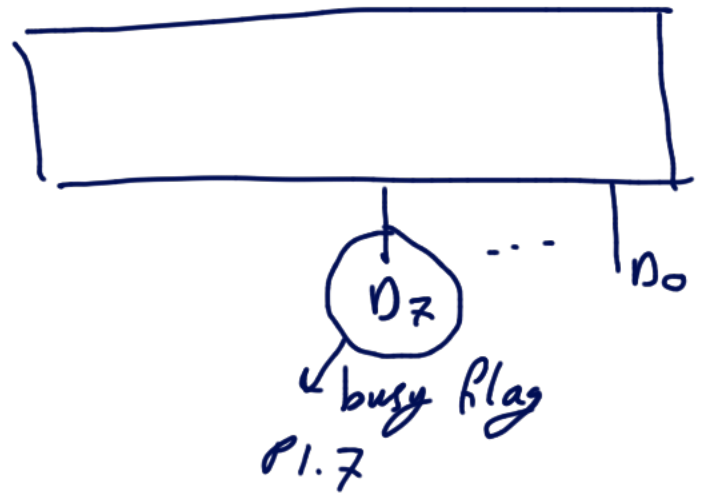
END

There are two ways to send data/command to LCD (9)

① using delay

② using D7 (busy flag)

Ready



READY: SETB P1.7

CLR RS

SETB RW ; receive from LCD

BACK: CLR E

ACALL DELAY2

SETB E

JB P1.7, BACK

RET

write subroutine to check whether the LCD ready or busy

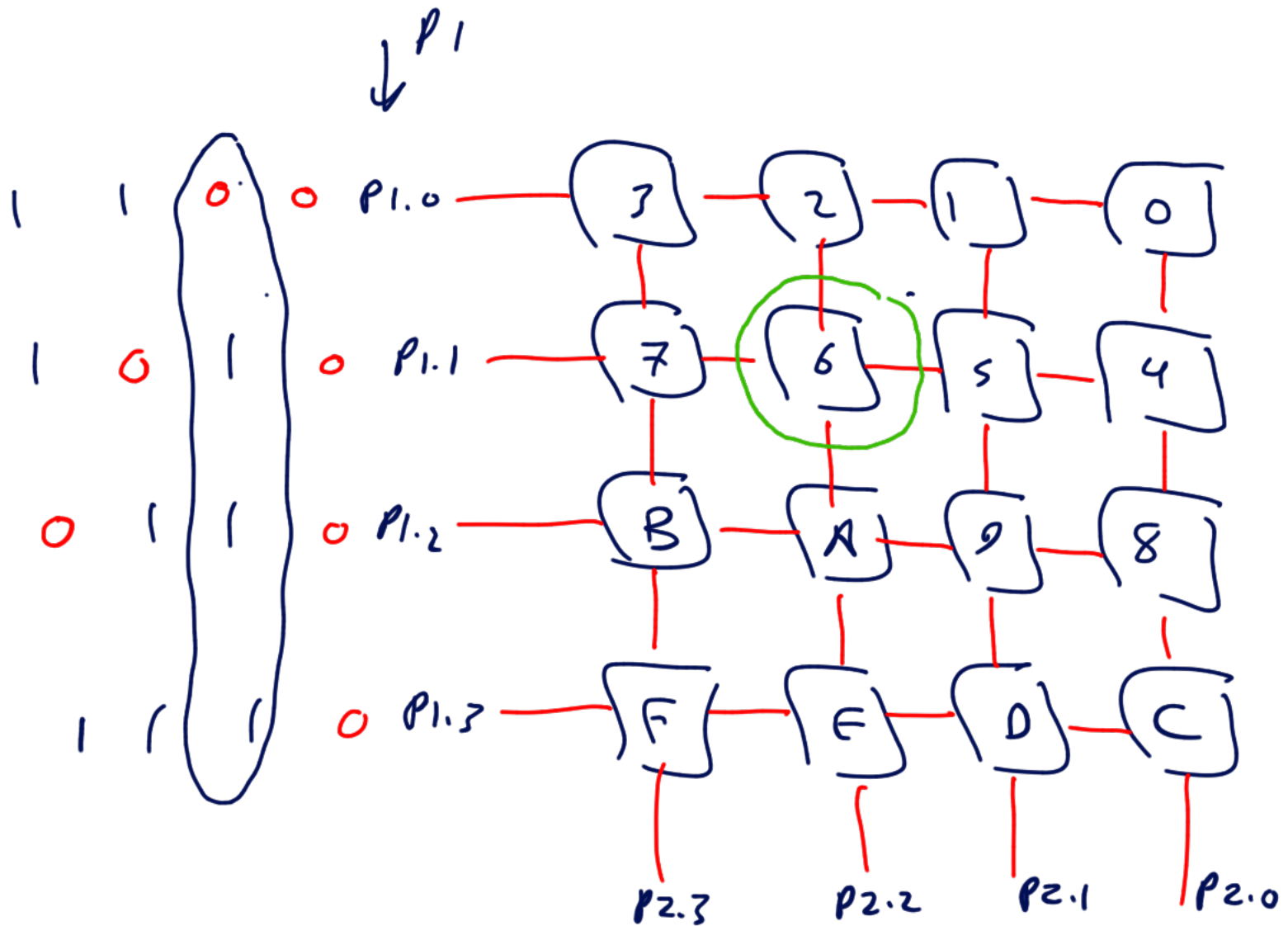
Given → Explain the function of this code?

Exam 2:

RET

Keyboard interface

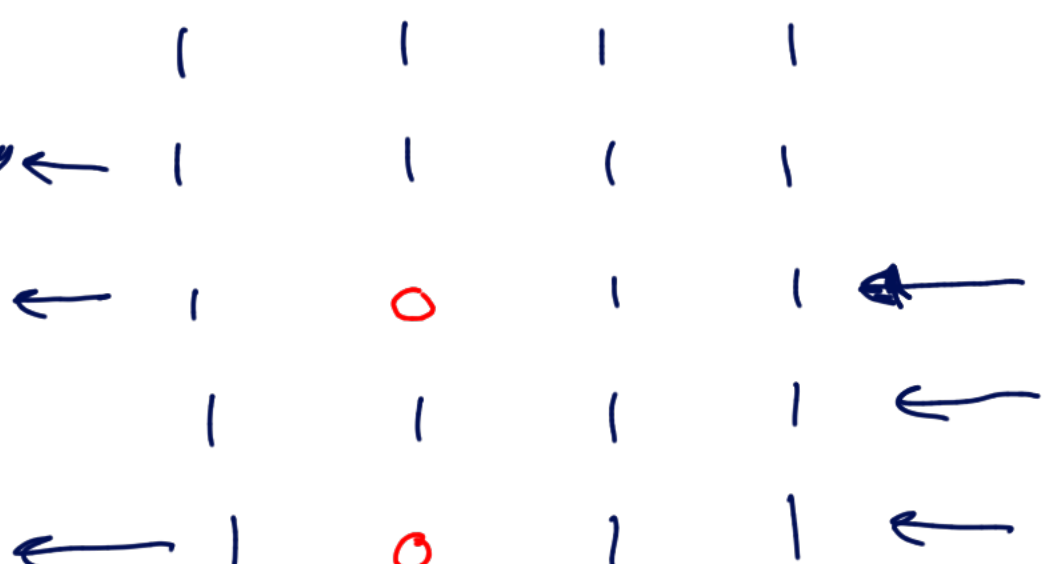
(10)



no pressed key ←

pressed key ←

is pressed [6] ←



(a) $D_3 - D_0 = 1110 \rightarrow \text{row}$
 $D_3 - D_0 = 1011 \rightarrow \text{column}$

key
(2)

(11)

(b) (7)

program

```

MOV P2, #0FFH
K1 : MOV P1, #00H
    MOV A, P2
    ANL A, #00001111B
    CJNE A, #00001111B, K1
K2 : ACALL DELAY
    MOV A, P2
    ANL A, #00001111B
    CJNE A, #00001111B, OVER
    SJMP K2
OVER :
```

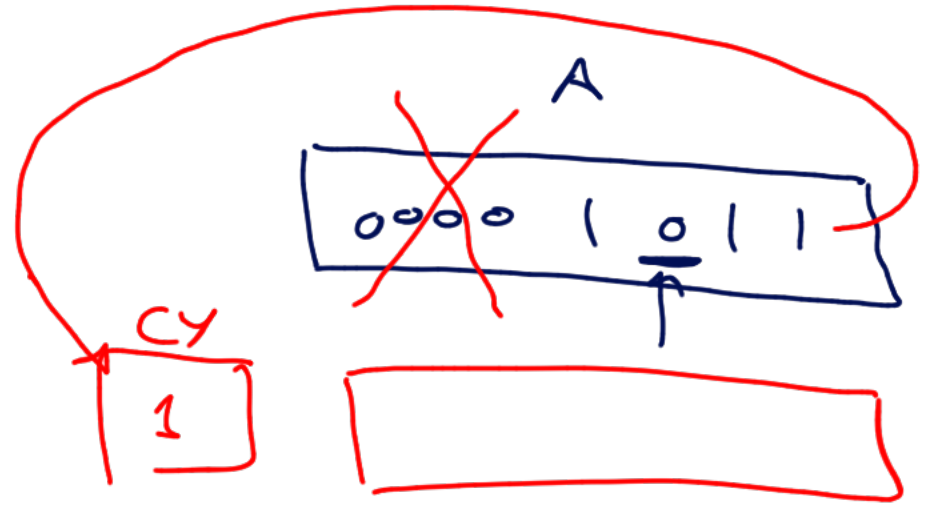
pressed key
To be sure that
all keys are
released.

detect
any pressed
key

OVER1

to be sure
that there is
a pressed key

Row1: mov DPTR, #KCODE1



KCODE1: DB '4', '5', '6', '7'

DPTR

A diagram showing the DPTR register pointing to the memory location of KCODE1, which contains the characters '4', '5', '6', and '7'. A red arrow points from the DPTR label to the first character '4'.

subroutine to detect (define) pressed key

Find: =

|||||

|||||

|||||

RRC